CUSTOMER SHOPPING BEHAVIOR ANALYSIS

An End-to-End Data Analytics Project

Python • SQL • Power BI

Prepared for Strategic Business Review
Dataset: 3,900 Transactions | 18 Variables | 4 Product Categories

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1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Attribute	Details
Number of Records	3,900 transactions
Number of Columns	18 variables
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
Missing Data	37 missing values identified in the Review Rating column

3. Exploratory Data Analysis (Python)

Data preparation and cleaning were carried out in Python using pandas, following a structured workflow:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used df.info() to review structure and .describe() to generate summary statistics.
- Missing Data Handling: Identified null values and imputed missing entries in the Review Rating column using the median rating for each product category.
- Column Standardization: Renamed columns to snake_case for improved readability and documentation.
- Feature Engineering: Created an age_group column by binning customer ages, and a purchase_frequency_days column derived from purchase data.
- Data Consistency Check: Verified redundancy between discount_applied and promo_code_used; the redundant promo_code_used column was dropped.
- Database Integration: Connected the Python script to PostgreSQL and loaded the cleaned dataset for downstream SQL analysis.

Summary statistics (selected fields):

Metric	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
Mean	44.07	59.76	3.75	25.35

Metric	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
Std. Dev.	15.21	23.69	0.72	14.45
Min	18	20	2.5	1
25th Percentile	31	39	3.1	13
Median (50%)	44	60	3.8	25
75th Percentile	57	81	4.4	38
Max	70	100	5.0	50

4. Business Analysis (SQL)

Structured analysis was performed in PostgreSQL to address ten key business questions, summarized below.

4.1 Revenue by Gender

Total revenue generated was compared between male and female customers.

Gender	Total Revenue (USD)
Female	75,191
Male	157,890

Male customers contributed roughly twice the revenue of female customers, despite a comparable average spend per transaction.

4.2 High-Spending Discount Users

Customers who applied a discount but still spent above the overall average purchase amount ($59.76) were identified, returning 839 customer records. A sample is shown below.

Customer ID	Purchase Amount (USD)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

839 customers used a discount while still spending above the average — indicating a meaningful segment of price-conscious but high-value shoppers.

4.3 Top 5 Products by Average Rating

Item Purchased	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4.4 Shipping Type Comparison

Average purchase amounts were compared between Standard and Express shipping options.

Shipping Type	Average Purchase Amount (USD)
Standard	58.46
Express	60.48

Express shipping customers spend slightly more on average, suggesting an opportunity to promote Express options to higher-value shoppers.

4.5 Subscribers vs. Non-Subscribers

Subscription Status	Total Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	59.49	62,645
No	2,847	59.87	170,436

While average spend is nearly identical between groups, non-subscribers represent the majority of total revenue simply due to their larger volume — highlighting significant headroom for subscription growth.

4.6 Discount-Dependent Products

The five products with the highest percentage of discounted purchases were identified.

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

4.7 Customer Segmentation

Customers were classified into New, Returning, and Loyal segments based on purchase history.

Customer Segment	Number of Customers
Loyal	3,116
Returning	701
New	83

The vast majority of customers (80%) fall into the Loyal segment, indicating strong overall retention but a small pipeline of new customer acquisition.

4.8 Top 3 Products per Category

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

4.9 Repeat Buyers and Subscriptions

Customers with more than five purchases were analyzed to assess whether repeat buying behavior correlates with subscription status.

Subscription Status	Repeat Buyers (>5 purchases)
No	2,518
Yes	958

Repeat purchasing is common among non-subscribers as well, suggesting that subscription status alone does not strongly predict purchase frequency — and that non-subscribed repeat buyers are prime conversion targets.

4.10 Revenue by Age Group

Age Group	Total Revenue (USD)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

Revenue is fairly evenly distributed across age groups, with Young Adults contributing the largest single share.

5. Power BI Dashboard

An interactive Power BI dashboard was developed to present these insights visually, enabling stakeholders to filter and explore the data by subscription status, gender, category, and shipping type. Key headline metrics displayed on the dashboard include:

Key Metric	Value
Number of Customers	3.9K
Average Purchase Amount	$59.76
Average Review Rating	3.75
Customers with Subscriptions	27%

The dashboard also includes interactive visuals for Revenue by Category, Sales by Category, Revenue by Age Group, and Sales by Age Group, along with filter panels for Subscription Status, Gender, Category, and Shipping Type.

6. Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers to convert the large base of high-spending non-subscribers, who currently generate the majority of revenue but show similar average spend to subscribers.

Customer Loyalty Programs

Introduce structured loyalty rewards to convert Returning customers (701) into the Loyal segment, and to deepen engagement among repeat buyers who are not yet subscribed.

Review Discount Policy

Reassess discount strategies for high-dependency products such as Hats, Sneakers, Coats, Sweaters, and Pants, balancing sales volume against margin impact.

Product Positioning

Feature top-rated items (Gloves, Sandals, Boots, Hat, Skirt) and best-selling products per category, such as Jewelry, Blouses, Sandals, and Jackets, in marketing campaigns.

Targeted Marketing

Focus marketing investment on high-revenue age groups, particularly Young Adults, and on Express-shipping customers, who demonstrate higher average spend.